Vector XSW

Equations and modal integration

A compact record of our theory discussion

1. Equation and ansatz

Geometry and conventions

Isotropic, nonmagnetic material: $\varepsilon(x,z) = 1+\chi(x,z)$. Period D along x; invariant along y; piecewise constant along z. Depth z increases into the sample. Lengths are in nm.

$$e^{-\mathrm{i}\omega t}, \qquad k_0 = \frac{2\pi}{\lambda}, \qquad \mathbf{h} = Z_0\mathbf{H}. \tag{1}$$

Vector wave equation

$$\boxed{\nabla\times\nabla\times\mathbf{E} = k_0^2\varepsilon\mathbf{E}}, \qquad \nabla\cdot(\varepsilon\mathbf{E}) = 0. \tag{2}$$

The code starts from the equivalent coupled Maxwell equations:

$$\nabla\times\mathbf{E} = \mathrm{i}k_0\mathbf{h}, \qquad \nabla\times\mathbf{h} = -\mathrm{i}k_0\varepsilon\mathbf{E}. \tag{3}$$

Relation to the scalar approximation

$$\nabla\times\nabla\times\mathbf{E} = \nabla(\nabla\cdot\mathbf{E}) - \nabla^2\mathbf{E}. \tag{4}$$

Only within a uniform material does $\nabla\cdot\mathbf{E} = 0$ follow directly. The code does not drop the divergence term globally or assume weak contrast. EUV wavelength alone does not establish the accuracy of a scalar approximation.

Bloch ansatz: Nikolaev-like notation

$$\boxed{\mathbf{E}(\mathbf{r}) = \sum_h \mathbf{E}_h(z)e^{\mathrm{i}\mathbf{k}_{h\parallel}\cdot\mathbf{r}_\parallel}}, \qquad \mathbf{k}_{h\parallel} = \left(k_x + \frac{2\pi h}{D},\, k_y\right). \tag{5}$$

$$\mathbf{E}_h = \begin{pmatrix} E_{x,h} \\ E_{y,h} \\ E_{z,h} \end{pmatrix}, \qquad \chi(x,z) = \sum_h \chi_h(z)e^{\mathrm{i}2\pi hx/D}. \tag{6}$$

Numerically, $h = -M,\ldots,M$: $N = 2M+1$ spatial orders. In each layer, the depth dependence is a sum of forward and backward modes:

$$\mathbf{E}(x,y,z) = e^{\mathrm{i}k_y y}\sum_n \left[T_n\mathbf{u}_n^+(x)e^{\mathrm{i}k_{z,n}z} + R_n\mathbf{u}_n^-(x)e^{-\mathrm{i}k_{z,n}z}\right]. \tag{7}$$

Here z is measured from a common reference plane in the layer. The physical mode profiles $\mathbf{u}_n^\pm$ contain all three electric components.

2. Vector eigenproblem

One layer at a time

Let $[f]_{hg} = f_{h-g}$ denote a Fourier convolution matrix. Define

$$C = [\varepsilon], \quad C_{hg} = \delta_{hg} + \chi_{h-g}, \qquad B = [1/\varepsilon], \quad A = B^{-1}, \tag{8}$$

$$X = \operatorname{diag}(k_{x,h}/k_0), \qquad Y = (k_y/k_0)I. \tag{9}$$

A implements the inverse Fourier factorization of the normal displacement. It is not Nikolaev's scalar eigenproblem matrix.

Eliminating longitudinal components gives

$$\frac{\mathrm{d}\mathbf{e}_t}{\mathrm{d}z} = \mathrm{i}k_0 P\mathbf{h}_t, \qquad \frac{\mathrm{d}\mathbf{h}_t}{\mathrm{d}z} = \mathrm{i}k_0 Q\mathbf{e}_t, \qquad \mathbf{e}_t = \begin{pmatrix}\{E_{x,h}\}\\ \{E_{y,h}\}\end{pmatrix}. \tag{10}$$

$$P = \begin{pmatrix} XC^{-1}Y & I - XC^{-1}X \\ YC^{-1}Y - I & -YC^{-1}X \end{pmatrix}, \tag{11}$$

$$Q = \begin{pmatrix} -XY & X^2 - C \\ A - Y^2 & YX \end{pmatrix}. \tag{12}$$

For $\mathbf{e}_t = \mathbf{w}_n e^{\mathrm{i}k_{z,n}z}$:

$$\boxed{\left(k_0^2 PQ - k_{z,n}^2 I\right)\mathbf{w}_n = 0}, \qquad q_n = k_{z,n}/k_0. \tag{13}$$

This is the vector counterpart of Nikolaev's scalar eigenproblem. There are $2N$ forward modes, with corresponding backward modes.

$$W = (\mathbf{w}_1, \ldots, \mathbf{w}_{2N}), \qquad V = QW\operatorname{diag}(q_n^{-1}). \tag{14}$$

W and V contain tangential electric and magnetic eigenvectors, respectively. Forward branches have $\operatorname{Im} q_n \geq 0$; propagating lossless branches have $\operatorname{Re} q_n > 0$.

Longitudinal fields and physical reconstruction

$$E_z = C^{-1}(Yh_x - Xh_y), \qquad h_z = XE_y - YE_x. \tag{15}$$

For backward modes, E_z, h_x, h_y change sign; E_x, E_y, h_z do not. The code reconstructs D_x/ε_0 from coefficients AE_x, then divides by the local $\varepsilon(x)$ to obtain physical $E_x(x)$.

Both s and p use this same eigenproblem. Only the incident excitation changes. Conical-incidence polarization coupling is retained.

Notation: P, Q here are Maxwell operators. They are not Nikolaev's interface basis and propagation matrix, denoted $\mathcal{P}, \mathcal{Q}$ below.

3. Boundary conditions and roughness

Sharp interface, no surface currents

$$\boxed{E_x^{(j)} = E_x^{(j+1)}, \quad E_y^{(j)} = E_y^{(j+1)}, \quad H_x^{(j)} = H_x^{(j+1)}, \quad H_y^{(j)} = H_y^{(j+1)}}. \tag{16}$$

At the same interface plane:

$$\boxed{\mathcal{P}_j \begin{pmatrix} \mathbf{T}_j \\ \mathbf{R}_j \end{pmatrix} = \mathcal{P}_{j+1} \begin{pmatrix} \mathbf{T}_{j+1} \\ \mathbf{R}_{j+1} \end{pmatrix}}, \qquad \mathcal{P}_j = \begin{pmatrix} W_j & W_j \\ V_j & -V_j \end{pmatrix}. \tag{17}$$

$$\mathcal{V}_{j,j+1} = \mathcal{P}_j^{-1} \mathcal{P}_{j+1}. \tag{18}$$

Prescribe the incident vector in the ambient and zero incoming amplitude from the substrate. These close the system for all reflected and transmitted modes.

Propagation through a depth increment d in our positive-depth convention is

$$\mathcal{Q}(d) = \operatorname{diag}\left(e^{\mathrm{i}k_0 q_n d}, e^{-\mathrm{i}k_0 q_n d}\right). \tag{19}$$

In computation, forward amplitudes are anchored at layer tops and backward amplitudes at bottoms; traversal uses only $e^{\mathrm{i}k_0 q_n d}$.

Interface roughness: effective modal extension

$$\kappa_{j,a} = \begin{cases} +k_0 q_{j,n} & \text{forward,} \\ -k_0 q_{j,n} & \text{backward,} \end{cases} \qquad F_{ab} = \exp\left[-\frac{\sigma^2}{2}(\kappa_{j,a} - \kappa_{j+1,b})^2\right]. \tag{20}$$

$$\boxed{\mathcal{V}_\sigma = \mathcal{V} \odot F}. \tag{21}$$

$\odot$ is elementwise multiplication. F is generally dense ($4N \times 4N$), not diagonal and not a matrix exponential. At $\sigma = 0$, every entry is one. Use complex signed wavevectors, without conjugation.

For a homogeneous interface this recovers

$$r_\sigma = r_0 e^{-2k_{z,j}k_{z,j+1}\sigma^2}, \qquad t_\sigma = t_0 e^{(k_{z,j}-k_{z,j+1})^2\sigma^2/2}. \tag{22}$$

r, t are tangential electric amplitudes. This corrects matching, not propagation. `Layer.sigma` applies at its top; `Structure.substrate_sigma` at the substrate top. Both default to 0 nm.

Sidewalls: independent Fourier damping

$$\Delta\varepsilon_m \mapsto \Delta\varepsilon_m e^{-G_m^2\sigma_{\mathrm{box}}^2/2}, \qquad G_m = 2\pi m/D. \tag{23}$$

Apply the same factor to each box's reciprocal-permittivity contrast. The zero harmonic is unchanged; `Box.sigma` defaults to 0 nm.

Scope: roughness is effective, with nominal material regions and no diffuse field. With interface roughness, $1 - R - T$ is not pure material absorption and need not be nonnegative.

4. Semi-analytic modal integration

Total electric intensity

For one layer of thickness d, combine forward and backward modes into index a. Using a common local reference $z = 0$ at the layer top, at $y = 0$:

$$\mathbf{E}(x, z) = \sum_a c_a \mathbf{u}_a(x) e^{\mathrm{i}\kappa_a z}, \qquad \kappa_a = \pm k_{z,n}. \tag{24}$$

$\mathbf{u}_a(x)$ is the reconstructed physical electric mode profile. For isotropic excitation, both s and p use

$$|\mathbf{E}|^2 = |E_x|^2 + |E_y|^2 + |E_z|^2. \tag{25}$$

For fluorescent-atom weight $n_f(x)$ constant through the layer depth:

$$\boxed{Y_j \propto \int_0^d \mathrm{d}z \int_{\text{period}} \mathrm{d}x\, n_f(x) |\mathbf{E}(x, z)|^2 = \sum_{a,b} c_a c_b^* L_{ab} U_{ab}}. \tag{26}$$

Lateral overlap

$$L_{ab} = \int_{\text{period}} n_f(x)\, \mathbf{u}_a(x) \cdot \mathbf{u}_b^*(x)\, \mathrm{d}x. \tag{27}$$

The dot product sums x, y, z electric components. Box intervals permit analytic integration of each pair of lateral Fourier harmonics.

Depth integral

$$\boxed{U_{ab} = \int_0^d e^{\mathrm{i}(\kappa_a - \kappa_b^*)z}\, \mathrm{d}z = \frac{e^{\mathrm{i}(\kappa_a - \kappa_b^*)d} - 1}{\mathrm{i}(\kappa_a - \kappa_b^*)}}. \tag{28}$$

When the denominator vanishes, $U_{ab} = d$. All mode-pair and forward/backward interference terms remain. Sum Y_j over finite layers.

Stable form used in the code

Each amplitude is instead anchored at z_a: layer top for forward modes, layer bottom for backward modes. The corresponding depth matrix is

$$\widetilde{U}_{ab} = \int_{z_{\text{top}}}^{z_{\text{bottom}}} e^{\mathrm{i}\kappa_a (z - z_a) - \mathrm{i}\kappa_b^* (z - z_b)} e^{-\mu z}\, \mathrm{d}z. \tag{29}$$

Use the anchored amplitudes with $\widetilde{U}$. The optional μ is a constant escape-attenuation coefficient; $\mu = 0$ gives the unweighted integral. The code evaluates exponentials from a bounded endpoint and uses finite zero-exponent limits.

For fixed photon energy and one emission line, angle-independent atomic prefactors cancel in a normalized curve. Keep spatial weights and any angle-dependent attenuation or geometry factors. Incident $|\mathbf{E}_{\text{inc}}| = 1$.

References and relation to the code

Nikolaev et al., arXiv:1908.11452v1: scalar notation, interface formalism, and semi-analytic fluorescence. The coupled vector eigenproblem and modal roughness extension above describe this package; they are not claimed as derivations from that scalar preprint.

Haase et al. (2016), Eq. 3: complex Névot–Croce reflection/transmission convention.

Implementation: `src/xsw/solver.py`, `observables.py`, `model.py`. Further detail: `docs/formulation.md`.